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RESEARCH

Bone Weathering Variability in Semiarid Environments: Insights from Taphonomic Experiments in the Dry Chaco, Central Argentina

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Abstract

Understanding the factors that govern bone weathering is essential for interpreting faunal preservation and site formation processes in archaeological contexts. This study presents the first four-years of an actualistic long-term experiment conducted in three contrasting depositional settings of the Dry Chaco ecoregion of Central Argentina: two open-air contexts (Arid Chaco and Mountain Chaco; $n=54$, $n=76$) and a sheltered context within the Mountain Chaco ($n=44$). Annual observations of weathering progression were recorded for camelid skeletal elements and integrated with climatic data to evaluate the influence of environmental and anatomical variables. Using a Bayesian regression model, we assessed the relative contributions of temperature, precipitation, solar radiation, and microenvironmental conditions to weathering trajectories. The model outputs demonstrate pronounced differences across contexts: rockshelter context showed an extremely low probability of transitioning into higher weathering stages, remaining at stage 0 across all years, while open-air settings exhibited faster and more heterogeneous weathering. Element morphology played a major role, with flat and axial bones showing higher susceptibility than long or compact elements. Microenvironmental factors—including shading, substrate, moisture regimes, and vegetation cover—proved more influential than broad regional climate, underscoring the need to interpret weathering within local ecological and geomorphological conditions. To explore ontogenetic effects, we generated exploratory estimates of neonatal weathering stages using adult–newborn differentials from an independent experiment. These suggest reduced survivorship of neonatal elements in open-air settings, highlighting the importance of age in mortality and taphonomic interpretations. Overall, our findings caution against using weathering stages as direct chronological markers and demonstrate the value of Bayesian approaches for modeling taphonomic processes. The methodological framework presented here provides a robust basis for expanding actualistic studies and improving archaeological inference in semiarid environments.

Keywords Taphonomy · *Lama* · South American Arid Diagonal · Córdoba · Open-air · Rockshelter

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Introduction

Bone weathering refers to the process by which the original microscopic organic and inorganic components of bone are progressively separated and destroyed through the action of physical and chemical agents operating *in situ*, both on the surface and within the soil (Behrensmeyer, 1978). To evaluate the impact of this taphonomic process on the formation of zooarchaeological assemblages, researchers commonly rely on Behrensmeyer's (1978) six-stage descriptive scale (0–5), developed from observations of mammal carcasses (> 5 kg) with known times and seasons of death in various environments—including swamp, dense woodland, open woodland, plains, bush, and lakebed—at Amboseli National Park, southern Kenya. This pioneering study demonstrated the potential of bone weathering as a valuable source of both ecological and taphonomic information.

Since its development, Behrensmeyer's scale has become a standard tool in zooarchaeological analysis and has been widely applied in paleontological, bioarchaeological and forensic research as well (Blau, 2017; Lyman, 1994; Stodder, 2018). In Argentina, it has been used to infer relative exposure time, depositional mode (attritional or catastrophic), assemblage integrity, and the degree of association among skeletal elements (*e.g.*, Álvarez, 2012; Bonomo, 2005; Gutiérrez, 2006; L'Heureux & Borrero, 2002; Massigoge, 2009; Medina, 2008; Mengoni Goñalons, 1999). Todd's (1987) modified version, which incorporates changes on articular surfaces, has also been widely applied (Costa, 2015, 2016, 2019; Izeta, 2007; Izeta & Scattolin, 2006; Izeta et al., 2009).

Despite its extensive use, several aspects of the method warrant reconsideration. As highlighted by Behrensmeyer (1978) and later by others (*e.g.*, Andrews & Cook, 1985; Lyman & Fox, 1989; Gutiérrez et al., 2016; Marín-Arroyo et al., 2023; Massigoge et al., 2010; Tappen, 1994), environmental conditions can accelerate or delay weathering, meaning that the relationship between stage and exposure time is not universal. In other words, bone weathering cannot be treated as a fixed chronological "clock" unless it is calibrated to local environmental conditions (Behrensmeyer & Miller, 2012). This principle underlies the long-term actualistic experiment presented in this paper.

Decades of taphonomic research have substantially advanced our understanding of bone weathering and its environmental variability. The rate and pattern of weathering are now known to depend on multiple intrinsic and extrinsic factors. Intrinsic factors include morphology, size, surface-to-volume ratio, bone structure, and mineral density (*e.g.*, Behrensmeyer, 1978; Behrensmeyer & Miller, 2012; González et al., 2012; Haynes & Wojtal, 2023; Madgwick & Mulville, 2012). These vary among species, skeletal elements, and ontogenetic stages, leading to non-uniform weathering responses (*e.g.*, Andrews, 1995; Cruz, 2014; Cunningham et al., 2011; Gutiérrez et al., 2016; Janjua & Rogers, 2008; Scheifler et al., 2020). Extrinsic factors include climate, sunlight exposure, depositional context, substrate type, and vegetation cover (*e.g.*, Andrews & Cook, 1985; Borrero, 1990; Pokines et al., 2018; Prassack, 2011; Ross & Cunningham, 2011; Tappen, 1994; Tuross et al., 1989). Subsurface weathering also occurs but at a much slower rate, and few studies have

systematically examined this condition (Lyman & Fox, 1989; but see Andrews & Armour-Chelu, 1998; Rapson & Todd, 1999; White & Hannus, 1983).

Given the many factors that influence bone weathering—and consequently affect the integrity and resolution of zooarchaeological assemblages—our research aims to calibrate this process in semiarid regions through a long-term actualistic experiment conducted in northwestern Córdoba Province, Central Argentina. Using camelid (*Lama* sp.) bones—ubiquitous in archaeological sites across Central Argentina and South America—the study investigates weathering dynamics across two subregions of the Dry Chaco ecoregion: the Mountain Chaco and the Arid Chaco, considering the distinct extrinsic factors operating in each. Within the Mountain Chaco, it also examines how weathering progresses under contrasting microenvironmental conditions (open-air vs. rockshelter settings) and compares the effects of this process on different skeletal elements.

Through a regionally novel analytical approach, the study explicitly evaluates the following hypotheses: (H1) weathering intensity increases with exposure time, leading to a greater likelihood of bones reaching more advanced weathering stages through time; (H2) regional climatic conditions influence weathering progression, such that variation in temperature, precipitation, and related environmental variables contributes to differences in weathering trajectories between the Arid Chaco and the Mountain Chaco; (H3) depositional context conditions weathering progression, with sheltered contexts limiting advancement to higher weathering stages relative to open-air settings; and (H4) skeletal element morphology influences weathering trajectories, such that flat and axial bones are more susceptible to advanced weathering than long and compact elements. Together, these hypotheses provide a structured basis for assessing how environmental, temporal, and anatomical factors shape bone preservation and the formation of zooarchaeological assemblages in semiarid environments.

Methodological Approach

Materials and Environmental Setting

The study consisted of three experiments (E1, E2, and E3) conducted at two locations in northwestern Córdoba Province, a semiarid region within the Dry Chaco, in Central Argentina (Fig. 1). Fresh bones from two adult llamas (*Lama glama*) were distributed in three wire mesh cages. The bones were acquired from Red Puna, a rural indigenous cooperative based in the high-altitude Andean plateau and ravines of Jujuy Province (Maina Waisman, 2022). The cages were lined with plastic woven sacks at the base to restrict access by people and animals, and to a lesser extent, vegetation growth. This setup followed some recommendations from the *Global Weathering Project* protocol (Marín-Arroyo et al., 2023). All bones retained soft tissues, including connective tissues (some of them were still articulated), periosteum, and flesh. Unlike a similar experiment carried out in the neighboring Pampean region, to the southeast, in this case the soft tissues were left intact rather than removed by boiling (González et al., 2012; Gutiérrez et al., 2016; Massigoge et al., 2010).

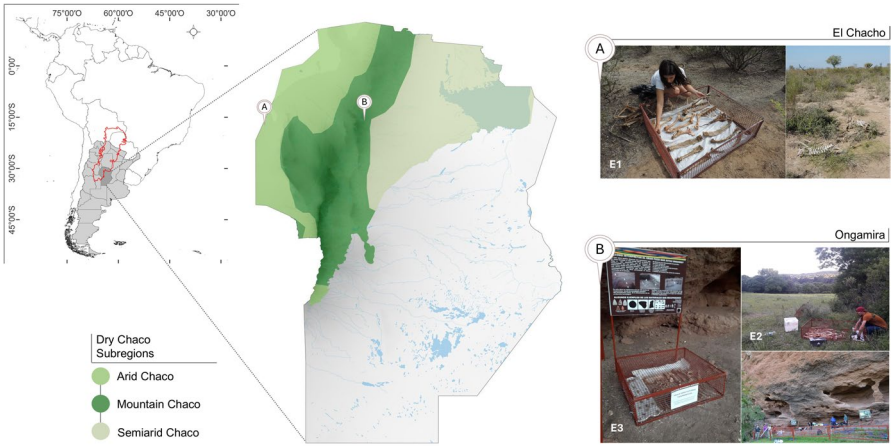

Fig. 1 Study areas in Córdoba Province, Argentina. The inset (top left) shows the location of Córdoba within Argentina and South America. In red, the Dry Chaco ecoregion is highlighted. The central map depicts the distribution of its subregions and the location of the two study areas: **A** El Chacho, in the Arid Chaco, and **B** Ongamira, in the Mountain Chaco. Photographs on the right illustrate these localities and the experiments conducted: E1, E2, and E3

The bones of both llamas were distributed as evenly as possible among the three cages, ensuring that at least one specimen of each skeletal element was present in every cage (Table 1). The only elements found in only two of the three cages (E1 and E2) are the sacrum and the caudal vertebrae. Apart from cranial and autopodial bones, all skeletal elements were present. Although there is yet no concrete archaeological evidence of the presence of domestic llamas in northwestern Córdoba Province, this camelid serves as an appropriate analogue for assessing the weathering process in its wild counterpart, the guanaco (*Lama guanicoe*), which was once abundant and widely distributed (Barri et al., 2023; Costa & Barri, 2018; Manzano-García et al., 2019).

The three experiments were conducted within the Dry Chaco ecoregion, encompassing two of its subregions: the Arid Chaco and the Mountain Chaco (Fig. 1). Experiment E1 was established in the Arid Chaco subregion, on a private field owned by the Farías family near El Chacho, a small rural settlement close to the Salinas Grandes saline depression. The landscape is dominated by open woodlands of *Aspidosperma quebracho-blanco* and *Prosopis nigra*, interspersed with perisaline shrublands composed of halophytic species capable of tolerating or regulating excess salt. These include small-leaved or leafless succulents with a distinctly xerophytic appearance, such as *Allenrolfea patagonica* and *Suaeda divaricata* (Zak & Cabido, 2002). Soils in this area exhibit high salinity, with frequent accumulations of carbonates, gypsum, and saline crusts (Rimondino, 2016). The climate is characterized by wide thermal amplitudes, both diurnal and seasonal, with average temperatures of approximately 25 °C in summer and 12 °C in winter. Precipitation is markedly seasonal, averaging around 130 mm in summer and less than 40 mm in winter.

Table 1 Number of elements of llama (*Lama glama*) used in each of the weathering experiments

Skeletal element	E1			E2			E3		
	Right	Left	Axial	Right	Left	Axial	Right	Left	Axial
Cervical vertebrae	-	-	1	-	-	-	-	-	2
Thoracic vertebrae	-	-	12	-	-	9	-	-	3
Lumbar vertebrae	-	-	7	-	-	4	-	-	3
Sacrum	-	-	1	-	-	1	-	-	-
Caudal vertebrae	-	-	6	-	-	9	-	-	-
Ribs	-	-	6	-	-	12	-	-	15
Scapula	-	1	-	1	1	-	1	-	-
Humerus	-	1	-	1	1	-	1	-	-
Radioulna	-	1	-	1	1	-	1	-	-
Carpals	-	7	-	7	7	-	7	-	-
Innominate	-	1	-	-	2	-	1	-	-
Femur	-	1	-	1	1	-	-	1	-
Patella	-	1	-	1	-	-	-	1	-
Tibia	-	1	-	1	1	-	1	-	-
Astragalus	-	1	-	1	1	-	1	-	-
Calcaneus	-	1	-	1	1	-	1	-	-
Tarsals	-	5	-	5	5	-	5	-	-
Total	54			76			44		

The other two experiments (E2 and E3) were carried out in the Mountain Chaco subregion, on land owned by the Supaga family in Ongamira, a well-studied archaeological area located in the northern sector of the Sierras Chicas, within the Sierras Pampeanas Australes (Cattáneo & Izeta, 2019a). One of these (E2) was placed in open-air conditions, approximately 40 m from the other (E3), which was located beneath the Deodoro Roca rockshelter (ADR, for its acronym in Spanish)—a well-documented archaeological site (Cattáneo & Izeta, 2019b; Cattáneo et al., 2013; Izeta et al., 2021; Menghin & González, 1954; Montes, 1943)—where bones were protected from direct climatic exposure. The Mountain Chaco subregion features a pronounced altitudinal vegetation gradient, ranging from mountain woodlands on lower slopes to shrublands and grasslands on higher elevations and plateaus (Zak & Cabido, 2002). Characteristic plant species include *Lithraea molleoides*, *Condalia buxifolia*, and *Zanthoxylum coco* (Robledo, 2021). Soils in the area are heterogeneous: E2 was placed on alluvial deposits typical of low-lying zones, whereas E3 rested on a harder, poorly developed substrate formed over the reddish sedimentary strata of the Saldán Formation (Centro Regional INTA Córdoba et al., 2025), providing contrasting microenvironmental conditions despite their close spatial proximity. The area has a temperate continental climate with warm, wet summers and cold, dry winters, during which snow may occasionally occur. Average temperatures range from 9 °C in winter to 25 °C in summer, with precipitation varying from about 10 mm in winter to 150 mm in summer.

Methods

Following their placement in November and December 2019, the cages at El Chacho and Ongamira were monitored annually. During each visit, observed bone surface modifications—*i.e.*, exfoliation, advanced exfoliation, mosaic cracking, bone loss, exposure of spongy tissue, and fibrous texture (Gutiérrez et al., 2016: 65)—were recorded. Each bone was then assigned a weathering stage based on the highest stage observed in at least 1 cm^2 of the element (Behrensmeyer, 1978:151). Additionally, the presence of insects and any loss of elements were documented.

For skeletal elements represented by multiple specimens within a single cage, the maximum observed weathering stage was recorded in order to capture the earliest onset of surface deterioration, a criterion commonly employed in taphonomic studies focused on preservation thresholds. As a complementary measure, the median and interquartile range (IQR) are reported. Median and IQR values, however, were not used as response variables in the statistical model (see below) because they may not correspond to discrete, observable weathering stages (*e.g.*, values falling between ordinal categories). Maximum observed stages were therefore retained for modeling, as they represent archaeologically meaningful thresholds while remaining compatible with ordinal regression frameworks and categorical graphical representations.

Concurrently, climatic conditions—specifically temperature, solar radiation, accumulated precipitation, and wind speed—were monitored using two weather stations. The Ongamira station was installed at the beginning of the experiment, approximately 300 m from the cage area, and its data are available on the Weather Underground platform (ID: IISCHI2). For El Chacho, climatic data were obtained from a station operated by the Provincial Administration of Hydric Resources (APRHI, for its acronym in Spanish) in Serrezuela (ID: APRHI MaayspCBA – Serrezuela), a village located about 30 km from the site.

The statistical analyses of climatic and taphonomic data were performed in the RStudio integrated development environment (R Core Team, 2024), with most figures generated using the *ggplot* package (Wickham, 2016). Climatic data from the Ongamira and Serrezuela stations were statistically compared using either a *t*-test or a Wilcoxon test, depending on data normality, which was assessed with the Shapiro–Wilk test. Following the approach of Madgwick and Mulville (2012), we evaluated the explanatory power of several variables in order to test our hypotheses: experiment (E1, E2, E3), depositional context (open-air, rockshelter), years of exposure (1, 2, 3, 4), element type (compact, long, flat, and vertebrae), and climate factors (mean temperature, accumulated precipitation, mean solar radiation, and mean wind speed). As an explanatory step, a classification tree (CHAID-type) was implemented using the *ctree()* function from the *partykit* package in R (Hothorn & Zeileis, 2015; Hothorn et al., 2006; Zeileis et al., 2008) to identify the variable explaining the largest proportion of variance in weathering stages and the order in which other variables entered the model.

Subsequently, an ordinal regression model was fitted using a Bayesian cumulative logit approach with the *brm()* function from the *brms* package (Bürkner, 2017, 2018, 2021), executed via *cmdstanr* (Gabry et al., 2025). Continuous climatic predictors (annual mean temperature, annual mean solar radiation, and annual accumulated

precipitation) were standardized prior to modeling. Mean wind speed was excluded due to collinearity, and "experiment" was removed to avoid redundancy with "context." The model used normal (0, 1) priors on fixed-effect coefficients and normal (0, 3) priors on the cumulative intercept thresholds. Coefficient stability was assessed through a posterior-based resampling procedure, in which datasets of different sizes ($n = 140$, 100, 75, and 50) were randomly drawn ten times each and refitted to evaluate the consistency of the posterior estimates. For each coefficient, posterior draws were extracted to compute 95% credible intervals; parameters whose intervals excluded zero were assigned + 1 or − 1 depending on the sign of the effect, and 0 otherwise. Net significance was then calculated by summing these scores across runs to identify predictors with consistent effects on bone weathering. To further summarize effect sizes from the Bayesian model, we computed odds ratios (OR) by exponentiating the posterior log-odds coefficients and extracted their 95% credible intervals. In addition, we calculated the probability of direction (pd) using the *bayestestR* package (Makowski et al., 2025), which quantifies the proportion of posterior samples supporting the sign of each effect. Both the database and the R Script to run the model are provided in Online Resource 1 and 2, respectively.

Results

Climatic Data

The statistical comparison of seasonal climatic data between Ongamira and Serrezuela revealed significant differences for most of the variables considered during the 2019–2024 period (Table 2). Serrezuela consistently exhibited higher maximum, mean, and minimum temperatures than Ongamira, with differences reaching statistical significance. In contrast, accumulated precipitation was substantially higher in Ongamira, which even recorded a snowfall in 2020, showing a mean difference of 231.3 mm. No statistically significant differences were observed for solar radiation or maximum wind speed between the two sites, although Ongamira displayed higher mean values for both variables. However, mean wind speed was higher in Serrezuela, and this difference was statistically significant.

Taphonomic Data

The main taphonomic modifications recorded across the three experimental settings are summarized below. Weathering progression is presented comparatively by deposition context, while detailed values appear in Table 3 and Figs. 2 and 3. Note that for elements represented by multiple specimens within a single cage (*e.g.*, ribs), the highest weathering stage recorded among them was used when compiling tables and figures. Online Resource 3 displays the complementary median and IQR metrics for each element by experiment as well as photographs of the main modifications described in the text.

Table 2 Seasonal comparison of climatic variables between Ongamira and Serrezuela (2019–2024). Mean values for each site are presented, along with the average difference between them, the 95% confidence interval, the results of the statistical test applied (paired *t*-test or Wilcoxon test, as appropriate), and the corresponding significance level. Statistically significant values (<0.05) are highlighted in bold

Climatic variable	Mean Ongamira	Mean Serrezuela	Mean difference	95% CI	*t*/V (gl)	*p*-value	Test applied
Maximum temperature (°C)	32.0	37.7	5.7	-	V = 1	0.002	Wilcoxon test
Mean temperature (°C)	15.5	22.6	–7.1	[–8.4; –5.8]	*t* = –12.2 (10)	< 0.001	*t*-test
Minimum temperature (°C)	1.2	7.1	–5.9	[–8.1; –3.6]	*t* = –5.8 (10)	< 0.001	*t*-test
Accumulated precipitation (mm)	399.8	168.6	231.3	[71.7; 390.8]	*t* = 3.2 (10)	0.0090	*t*-test
Solar radiation (Wt/m^2)	199.1	195.4	3.7	[1.5; 61.8]	*t* = 0.4 (10)	0.6713	*t*-test
Maximum wind speed (km/h)	31.7	27.0	4.7	-	V = 54	0.0673	Wilcoxon test
Mean wind speed (km/h)	2.8	7.9	5.1	-	V = 4.5	0.0127	Wilcoxon test

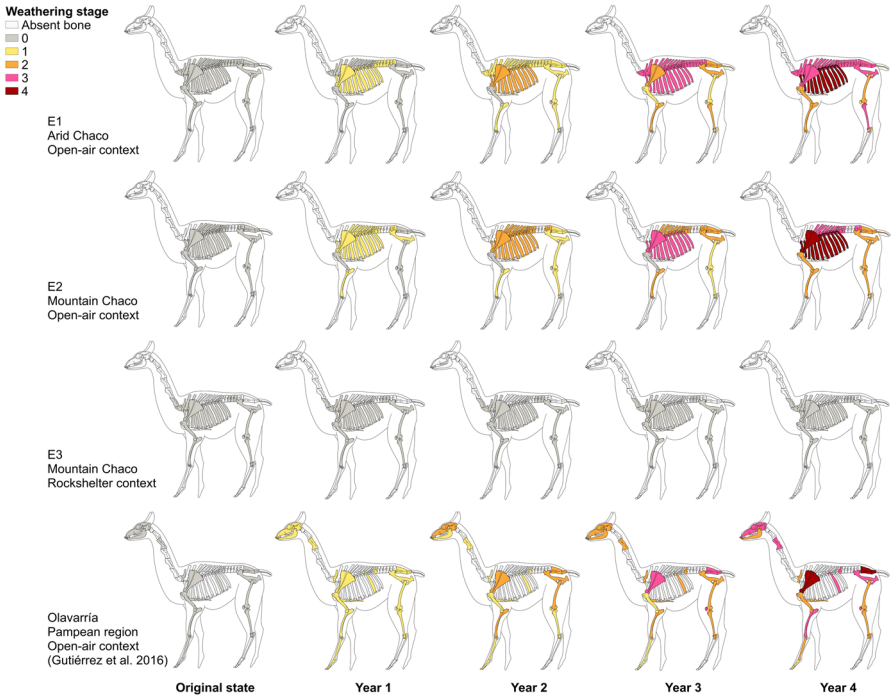

Fig. 2 Annual progression of weathering stages by skeletal element across the three experiments. Comparative data from the adult guanaco experiment reported by Gutiérrez et al., (2016: 64) in an open-air context of the Pampean region (Olavarría, Buenos Aires Province) are also included

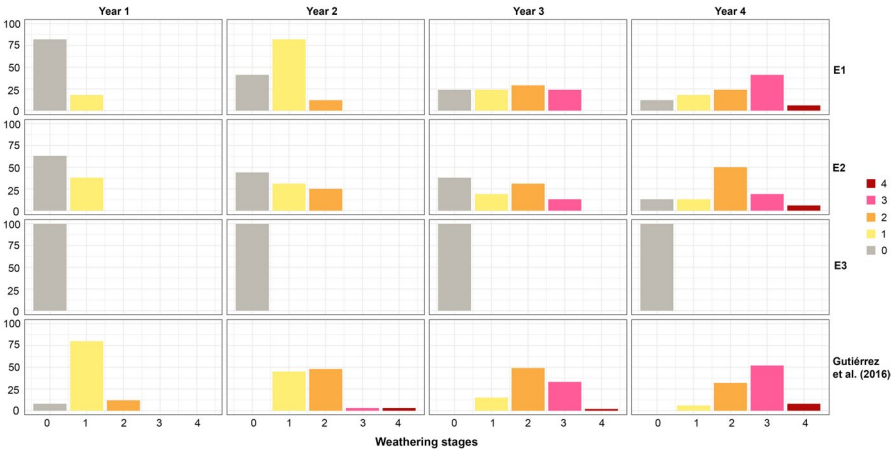

Fig. 3 Annual progression of weathering stages in the three experimental contexts in northwestern Córdoba Province. Comparative data from the adult guanaco experiment reported by Gutiérrez et al., (2016: 63) in an open-air context of the Pampean region (Olavarría, Buenos Aires Province) are also included

Table 3 Maximum weathering stages per element recorded annually across the three experiments

Skeletal element	E1					E2					E3				
	n	**1**	**2**	**3**	**4**	**n**	**1**	**2**	**3**	**4**	**n**	**1**	**2**	**3**	**4**
Cervical vertebrae	1	0	1	3	3	0	-	-	-	-	2	0	0	0	0
Thoracic vertebrae	12	0	1	3	3	9*	1	2	2	3	3	0	0	0	0
Lumbar vertebrae	7	1	1	3	3	4	1	2	2	3	3	0	0	0	0
Sacrum	1	0	1	2	3	1	0	1	2	2	0	-	-	-	-
Caudal vertebrae	6	0	0	1	1	9	0	0	0	2	0	-	-	-	-
Ribs	6	1	2	3	4	12	1	2	3	4	15	0	0	0	0
Scapula	1	1	2	2	3	2	1	2	3	3	1	0	0	0	0
Humerus	1	0	0	1	2	2	0	0	0	2	1	0	0	0	0
Radioulna	1	0	1	2	2	2	1	1	2	2	1	0	0	0	0
Carpals	7	0	0	0	1	14	0	0	0	0	7	0	0	0	0
Pelvis	1	0	1	2	3	2	1	1	2	2	1	0	0	0	0
Femur	1	0	1	1	2	2	0	1	1	2	1	0	0	0	0
Patella	1	0	0	0	1	1	0	0	0	1	1	-	-	-	-
Tibia	1	0	1	2	3	2	0	1	1	2	1	0	0	0	0
Astragalus	1	0	0	0	0	2	0	0	0	1	1	0	0	0	0
Calcaneus	1	0	0	1	2	2	0	0	1	2	1	0	0	0	0
Tarsals	5	0	0	0	0	10	0	0	0	0	5	0	0	0	0

*Only 7 thoracic vertebrae remained after the second year in E2

Weathering Under the Rockshelter (E3): Persistent Stage-0 Preservation

Throughout the four years of monitoring, all elements placed under the rockshelter remained at weathering stage 0. Bones consistently retained moisture and a greasy texture, with no evidence of exfoliation, cracking, or bone loss. A progressively thicker layer of sediment and sandstone particles, derived from the rockshelter wall, accumulated on the assemblage, sometimes accompanied by small clasts on flat bones. Soft tissues persisted longer in this context than in any other, and no disarticulation was observed. Dermestid larvae and, occasionally, green mold remained associated with some elements, but these did not result in visible bone alteration. Overall, the sheltered assemblage exhibited remarkable stability, showing no progression in weathering despite long-term exposure.

Weathering in Open-Air Contexts (E1 and E2): Progressive Surface Modification

In contrast to E3, both open-air assemblages (E1 and E2) exhibited clear and progressive weathering. During the first two years, most elements darkened and showed fat residues, with an increasing representation of stages 1 and 2, particularly in axial elements. Scapulae, ribs, and lumbar vertebrae developed exfoliation and localized bone loss. In E2, a gastropod (Gastropoda *cf. Plagiodontes daedalus*) was recorded attached to the pelvis, and annual vegetation growth frequently covered the cage. Furthermore, a small carnivore entered the E2 cage during the second year,

removing seven articulated thoracic vertebrae. Five were recovered a few meters away, showing gnawing marks and partial destruction of dorsal processes.

From the third year onward, weathering intensified in both contexts, with a balanced distribution of stages 0 to 3 and advanced alterations predominantly affecting axial bones, scapulae, and some girdle and zeugopod elements. Flat bones developed fibrous texture, advanced exfoliation, and bone loss, while long bones showed widespread longitudinal cracking and exfoliation, occasionally accompanied by mosaic cracking on articular surfaces. Soft tissues decreased markedly in both open-air assemblages over time, remaining only as dry ligaments and tendons by the third year, which maintained limited articulation in elements such as vertebrae, patella, and femur. This condition was more pronounced in E1 than in E2. By the fourth year, stage 4 was recorded in ribs and scapulae in E2, whereas in E1 stage 3 predominated in vertebrae, girdles, and tibia. Only carpals, tarsals, and the astragalus remained largely unaffected. In general, despite the climatic differences recorded between the two sites (Table 1), weathering progression in E1 and E2 exhibited broadly similar patterns.

Multivariate Analysis

Figure 4 presents the classification tree (CHAID) obtained from the data collected in our experiments. The first node (root) indicates that the depositional context accounts for the largest proportion of variance in weathering stages ($p < 0.001$), clearly separating specimens exposed in the open air from those under the rockshelter. Within each branch, the next most influential variable is "year" ($p < 0.001$ in the open-air branch), showing that exposure time—specifically the first two years versus the third and fourth years—strongly modulates weathering progression. Subsequently, element type repeatedly appears as a discriminating variable ($p < 0.001$ and $p < 0.05$ in different nodes), indicating that bone morphology (compact, long, flat, vertebrae) conditions the likelihood of reaching higher weathering stages. Terminal nodes display relative frequency histograms of the stages (0, 2, 4 in our dataset), with darker bars indicating higher stage values. For example, Node 15 ($n = 56$), corresponding to "rockshelter," contains only stage 0, illustrating the absence of weathering under this condition. In contrast, nodes associated with "open-air" and certain element types or years of exposure show a broader range of stages, indicating more advanced weathering. Overall, the tree indicates that depositional context, exposure time, and bone morphology are the primary factors structuring variation in weathering stages in our experiments.

To complement this exploratory analysis and assess the stability and direction of these effects, we fitted a Bayesian ordinal regression model. The results confirm and refine the pattern displayed in the classification tree: depositional context and bone morphology are the main predictors of weathering progression, followed by exposure time and certain climatic variables. Particularly, the net significance of the coefficients indicates a strong negative effect of the "rockshelter" context compared to "open-air" exposure, meaning that bones placed under shelter have a markedly lower probability of reaching advanced weathering stages. Among element types,

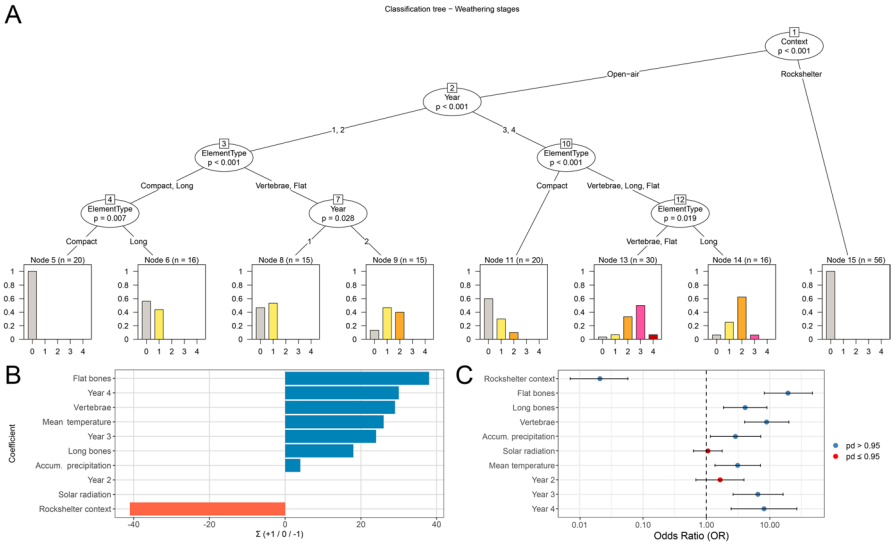

Fig. 4 **A** Classification tree (CHAID) for weathering stages. Relative-frequency histograms of stages 0–4 are shown in each terminal node; sample size (n) is indicated in parentheses. **B** Net significance of the coefficients in the Bayesian ordinal model. The vertical axis lists the model predictors, while the horizontal axis shows the cumulative score $\Sigma(+1/0/-1)$, where each run or subsample was assigned +1 if the 95% credible interval of the coefficient was > 0, −1 if it was < 0, and 0 if it included 0. **C** Forest plot of odds ratios (OR) for fixed effects in the Bayesian cumulative logit model. Points are colored by probability of direction (pd), and error bars represent 95% credible intervals. Note: Reference categories ("Open-air" for Context and "Year 1" for Year) are not displayed, as their effects are absorbed in the model intercepts, which were omitted for clarity (see Online Resource 4)

flat bones, vertebrae, and, to a lesser extent, long bones show consistently positive coefficients, confirming that morphology strongly conditions the probability of reaching advanced stages. As expected, longer exposure times—specifically the third and fourth years—display strong positive and stable effects, indicating a cumulative effect of duration on weathering. Lastly, regarding climatic variables, mean temperature emerges as a relevant positive factor, while accumulated precipitation and solar radiation show small or uncertain effects (credible intervals overlapping zero). Full model diagnostics are provided in Online Resource 4.

Additionally, Fig. 4C presents a forest plot of odds ratios (OR) with 95% credible intervals and associated probability of direction (pd). This visualization highlights the relative magnitude and directional support of each predictor. The strongly negative effect of the rockshelter context (OR = 0.02; pd = 1) shows an unambiguous reduction in the probability of reaching advanced weathering stages. Likewise, flat bones, vertebrae, and—to a lesser extent—long bones display consistently positive effects (all OR > 4; pd ≥ 0.999), indicating a high and stable likelihood of progressing to higher stages. Exposure duration also shows cumulative effects, with years 3 and 4 exhibiting strong, well-supported increases in weathering probability (OR > 6; pd ≈ 1), whereas year 2 presents only a weak trend (credible interval overlapping 1). In contrast, climatic predictors display wider uncertainty: temperature and precipitation show moderate positive effects with high pd values, while solar radiation

exhibits low directional support (pd ≈ 0.59) and a credible interval spanning 1, indicating considerable uncertainty about its contribution. Full numerical summaries are provided in Online Resource 4.

Taken together, these results indicate that morphological and contextual variables dominate the weathering process, while climate plays a secondary but detectable role.

Discussion

The results of our experiments add to a long series of contributions to the understanding of bone weathering processes developed worldwide since Behrensmeyer's (1978) pioneering work. Our results highlight strong variability in weathering trajectories within the Dry Chaco, driven not only by regional differences (Arid vs. Mountain Chaco) but also by fine-grained microenvironmental conditions and anatomical variation among skeletal elements. The main findings from the four-year monitoring period can be summarized as follows: (1) markedly slower weathering under rockshelters (E3); (2) progressive weathering with converging trajectories in open-air experiments (E1 and E2); (3) strong microenvironmental effects shaping weathering rates; (4) a critical threshold around the third year in open-air contexts; and (5) consistent intraskeletal variability, with flat and axial bones generally more affected than long and compact elements.

As has often been noted, weathering stages have been widely—and in many cases uncritically—used as a "taphonomic clock" to estimate exposure duration. However, numerous studies have cautioned that weathering rates may be affected by a wide range of intrinsic and extrinsic factors, including species, skeletal morphology, ontogenetic development, climate and microenvironmental conditions (*e.g.*, Alunni et al., 2017; Andrews, 1995; Andrews & Cook, 1985; Andrews & Armour-Chelu, 1998; Andrews & Whybrow, 2005; Behrensmeyer, 1978; Behrensmeyer et al., 1979; Borrero, 1990; Cruz, 2014; Gutiérrez et al., 2016; Haynes & Wojtal, 2023; Junod, 2013; Madgwick & Mulville, 2012; Marín-Arroyo, et al., 2023; Morales et al., 2024; Pokines et al., 2018; Pokines & Ames, 2015; Prassack, 2011; Ross & Cunningham, 2011; Scheifler et al., 2020; Tappen, 1994; White & Hannus, 1983). Our experiments reinforce these cautions, showing substantial variability across contexts and skeletal elements.

The most striking difference, qualitatively and quantitatively recorded during the first four years, was that between sheltered and open-air experiments. The stage-0 pattern recorded under the rockshelter (E3) parallels several instances of exceptional bone preservation described elsewhere: for example, Andrews and Cook's (1985) study of a juvenile cow carcass protected under an overhang surrounded by vegetation in South West England, which did not display visible weathering after more than seven years, or the outstanding preservation of paleontological material at Natural Trap Cave in Wyoming, USA (Wang & Martin, 1993). Regionally, similar conditions have been documented in the Mountain Chaco, where carnivore-accumulated assemblages recovered from caves in Characato exhibit weathering profiles

dominated by stages 0 and 1 (Weihmüller et al., 2021a; Brizuela, 2025; Cattáneo et al., 2016; Nóbile et al., 2025).

In contrast, the open-air contexts (E1 and E2) exhibited progressive weathering and greater stage variability, a pattern consistent with H1. Based on climatic differences between Ongamira and Serrezuela (Table 1), one might expect more accelerated weathering in E1 (Arid Chaco) relative to E2 (Mountain Chaco). However, E2 weathered slightly faster during the first two years, and the two assemblages converged thereafter, with increasing representation of stages 2 and 3 and the first appearance of stage 4 in the fourth year. Overall, long and compact bones proved more resistant, while flat and axial elements were most affected (Fig. 2).

This convergence is best explained by microenvironmental factors. In E1, shading provided by dense woody vegetation, particularly during summer, likely reduced direct exposure and slowed weathering (Alunni et al., 2017; Andrews & Cook, 1985; Behrensmeyer, 1978; Borrero, 1990; Tappen, 1994). This protective effect is especially notable given field observations indicating that bones exposed to full sunlight in the Arid Chaco tend to weather rapidly. In contrast, E2 experienced annual herbaceous growth that covered the cage, trapping dew moisture and producing diurnal wetting–drying cycles, while occasional snowfall likely added freezing–thawing cycles, both processes known to accelerate bone weathering (Junod, 2013; Pokines et al., 2016, 2018; Prassack, 2011). These results offer only partial support for H2: climatic variables do contribute to differences between the Arid and Mountain Chaco—especially mean temperature—but their effects are weaker than initially expected, suggesting that microenvironmental conditions often override broader regional trends, as anticipated by H3.

When compared to the original observations made by Behrensmeyer (1978) for the African savanna, some similarities and differences arise. Based on the monitoring of 35 carcasses with known times of death, the estimated weathering rate at Amboseli indicates that most carcasses exposed for three years or less display weathering stages 0, 1, or 2.[1] Carcasses exposed for more than three years show a wide range of weathering stages, but none fall within stages 0 or 1, allowing the author to distinguish—with relative confidence—carcasses exposed for less than three years (Behrensmeyer, 1978: 157). Our experiments also identify the third year as a key threshold for accelerated weathering in open-air contexts (Fig. 4). However, weathering appears to progress somewhat faster in the Dry Chaco, with the first stage-3 elements appearing during the third year. Moreover, in both E1 and E2, elements in stages 0 and 1 persisted through at least the fourth year (Fig. 3).

These differences may partly reflect contrasting initial conditions. Behrensmeyer examined whole carcasses protected by soft tissues, whereas our experiments used isolated, partially defleshed elements. Soft tissues delay the onset of weathering (Alunni et al., 2017; Borrero, 1990; Morales et al., 2024), and the time required for disarticulation depends on variables such as scavenger activity, climate, and substrate (Blumenschine, 1989; Borrero, 1990; Faith et al., 2007; Marchionni et al.,

[1] To assign a weathering stage to a carcass, the author adopts the highest stage recorded among its constituent elements (Behrensmeyer, 1978).

2020; Scheifler et al., 2020). A camera-trap study near E1 revealed that body size and scavenger competition affect the speed at which bones become exposed (Weihmüller et al., 2021b). Although different from Behrensmeyer's naturalistic context, the initial conditions of our bones may better approximate human-processed and discarded remains.

Subsequent work following Behrensmeyer's pioneering study emphasized the importance of accounting for intraskeletal variation in the expression of weathering (*e.g.*, Lyman, 1994; Lyman & Fox, 1989; Madgwick & Mulville, 2012; Massigoge et al., 2010), an aspect that Behrensmeyer did not explore in detail, likely because her focus was oriented toward broader paleoecological questions (Behrensmeyer, 1978). In zooarchaeological assemblages, which are typically composed of disarticulated, fragmented, and taphonomically averaged elements, recognizing that weathering progresses differentially across bone types is essential for interpreting their taphonomic history. Our results demonstrate strong anatomical differentiation, as predicted in H4: in open-air contexts, flat bones, vertebrae, and long bones weather more rapidly, which aligns with camelid studies from Patagonia and the Pampean region (Borrero, 2007; Cruz & Muñoz, 2010; Morales et al., 2024).

Experimental data from Gutiérrez et al. (2016) further contextualize these findings. Their adult guanaco weathered more rapidly than specimens in E1 and E2, particularly during the first two years, and no elements remained at stage 0 by year two—a pattern reminiscent of Amboseli. In contrast, our open-air experiments retained compact bones at stage 0 for longer (Figs. 2 and 3). Several factors likely contributed to rapid weathering in Olavarría: the bones had been previously boiled, while ours retained soft tissues and sometimes partial articulation (Alunni et al., 2017; Morales et al., 2024); and the sandy, vegetation-poor substrate probably accelerated drying and surface alteration (Gutiérrez et al., 2016; Massigoge et al., 2010). Still, by the third year weathering trajectories converged across contexts, with axial elements and scapulae reaching stages 2–3 and long bones remaining largely in stage 2 (Fig. 3).

Ontogenetic variation introduces another dimension. Gutiérrez et al. (2016) documented marked age-related differences: immature individuals weathered faster than adults, reaching stage 5 within eight years. The juvenile reached stage 5 by year three, and the newborn by year six; several neonatal elements—including the cranium, mandible, a cervical vertebra, and a rib—were completely destroyed once stage 5 was reached. Although our study did not incorporate ontogenetic variation directly, we generated exploratory estimates of neonatal weathering stages by applying adult–newborn differentials from Olavarría (Gutiérrez et al., 2016). These estimates (Online Resource 5) paralleled adult trajectories but suggested heightened vulnerability in neonates: E3 would preserve them at stage 0, whereas E1 and E2 would lead some elements to reach stages 3–4 by year four. As accurately noted in various studies, these patterns highlight the importance of considering ontogeny when interpreting mortality and taphonomic profiles (Behrensmeyer, 1978; Behrensmeyer et al., 2003; Fiorillo, 1989; Gutiérrez et al., 2016). Our estimation procedure, however, should be interpreted strictly as a heuristic exercise intended to illustrate potential ontogenetic vulnerability rather than provide empirical results. It assumes that adult–neonate differences remain constant through time and are comparable

across regions—assumptions unlikely to hold universally. Empirical neonatal experiments under similar environmental conditions will be essential for validating and refining these estimates and for detecting potential nonlinear or threshold responses in subadult weathering trajectories.

From an archaeological standpoint, the low weathering observed in E3 supports certain expectations: assemblages recovered from rockshelters within the Mountain Chaco should generally exhibit good preservation, especially when protected from direct weathering agents and provided that burial conditions are favorable (see White & Hannus, 1983). However, the high sedimentation rates documented in these settings (Zárate, 2019) increase the potential for mixing between archaeological and contemporary materials, particularly in the case of small, compact elements (*e.g.*, carpals, tarsals, teeth), which are more prone to vertical movement (Borrero, 2007; Otaola & Tripaldi, 2016). Larger elements, in contrast, may remain on the surface for longer periods due to the hardness of the substrate beneath rockshelters, exposing them to horizontal displacement by humans or animals (Otaola & Tripaldi, 2016).

In open-air contexts, continuous exposure to climatic and biotic agents, together with the low likelihood of burial in many settings, likely contribute to substraction (like the one recorded in E2 by a carnivore), high fragmentation and generally poor preservation of zooarchaeological remains documented at open-air archaeological sites (Caminoa, 2023; Medina & Rivero, 2020; Robledo, 2020). Despite climatic differences across the region, factors such as burial potential and soil characteristics may ultimately determine the survival of bones exposed to the elements. In the Arid Chaco, field observations indicate that rainfall and wind erosion produce cycles of burial and exposure that accelerate weathering and lead to bone destruction (Costa et al., 2022). In the Mountain Chaco, soils are more heterogeneous: low-lying valley areas exhibit high sedimentation rates, whereas upland zones dominated by bedrock show little sediment accumulation (Martino et al., 2020; Zárate, 2019). In this regard, the recently initiated experiment at the Quebrada del Condorito National Park —located in a highland grassland environment of the Sierras Grandes— is expected to provide complementary information on weathering trajectories in higher-altitude contexts than those examined here (Arias et al., 2024; Pía & Mondini, 2025).

Conclusions

The actualistic experiments presented here underscore the need for caution when interpreting bone weathering in archaeological contexts. Before inferring attritional processes, it is essential to evaluate weathering patterns through anatomically structured profiles and at the species level, as emphasized by previous work (Cruz & Muñoz, 2010; Madgwick & Mulville, 2012; Prassack, 2011). Our results show that weathering does not progress uniformly across skeletal elements, and that intraskeletal variability can substantially influence the apparent composition and taphonomic character of zooarchaeological assemblages.

The sheltered experiment (E3) demonstrates that good preservation within rockshelters does not necessarily imply rapid burial or deliberate refuse management, as has sometimes been proposed for archaeological sites in the region (*e.g.*, Costa, 2016). Instead, natural protection from sunlight, precipitation and wind —combined with localized sediment accumulation— can be sufficient to maintain bones in early weathering stages for several years (Kligmann et al., 2021). Thus, cultural explanations for low weathering should not be invoked without first considering the microenvironmental regimes of sheltered settings.

In contrast, our open-air experiments (E1 and E2) highlight the far lower likelihood of bone survival in exposed environments. Unless sedimentation rates are high enough to promote rapid burial, bones are unlikely to persist, particularly in areas subject to intense solar radiation or fluctuating moisture regimes. These findings reinforce the importance of situating weathering data within broader geomorphological and depositional frameworks, a point repeatedly emphasized in regional taphonomic studies (Alunni et al., 2017; Belardi & Carballo Marina, 2003; Borrero, 2007; López & Chiavazza, 2019; Marchionni et al., 2020, 2022; Otaola & Tripaldi, 2016; Scheifler et al., 2020). Ultimately, the geomorphic context (more than climate alone) may determine the long-term archaeological preservation of faunal remains in the Dry Chaco.

The Bayesian regression model employed in this study provides a flexible analytical framework for modeling weathering trajectories. Its structure allows new experimental datasets to be incorporated even when they differ in starting dates or duration, provided that annual climatic variables and weathering observations are available. The model can also be expanded to include additional predictors not considered here, such as species, age—two factors incorporated in the original formulation by Madgwick and Mulville (2012)—or specific microenvironmental conditions. Despite the limitations inherent in the variables included in the present analysis, one of the key strengths of Bayesian approaches lies in their adaptive nature: as sample sizes and dataset variability increase, models become progressively more robust, better calibrated and more informative (Buck & Litton, 1991; Etz et al., 2018).

Taken together, our results emphasize that bone weathering must be interpreted in relation to skeletal anatomy, depositional microenvironments and geomorphological context. Incorporating these dimensions is essential for generating reliable taphonomic and behavioral inferences in the archaeological record of the Dry Chaco and other semiarid environments.

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Author Contribution M.P.W., A.D.I. and G.R.C. designed and initiated the research. M.P.W. conducted data collection, performed the analyses, interpreted the results and lead the writing of the original draft. A.D.I. and G.R.C. supervised the project and reviewed the original draft.

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Data Availability The dataset used in this study is provided as supplementary material.

Declarations

Competing interests The authors declare no competing interests.

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